LiquidAI: Collective Intelligence Through Agent–Environment Coevolution

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Abstract

The dominant paradigm in artificial intelligence treats cognition as a property of isolated agents operating over passive environments. Yet biological intelligence emerged through continuous reciprocal interactions between organisms and the worlds they inhabit and transform. Across evolution, environments became progressively integrated into computation itself: storing information, coordinating behavior, constraining dynamics, and enabling increasingly complex forms of collective organization.

The **LiquidAI** project proposes a radically new framework for artificial intelligence grounded in populations of communicating agents embedded within evolving environments. We aim to study how collective cognition, ecosystem engineering, social learning, and evolutionary adaptation jointly generate new forms of individuality, ultrasocieties, and hybrid human–AI organizations.

Our approach spans scales and substrates, from engineered xenobots and anthrobots capable of modifying their environments, to digital ecosystems of large language model agents interacting within spatially extended virtual worlds. A central goal is to construct morphospaces of possible cognition capable of identifying unexplored regions in the landscape of intelligence and guiding the emergence of radically novel artificial systems.

LiquidAI seeks to establish a science of ecological intelligence in which cognition is understood not as a property of isolated entities, but as a distributed process emerging from closed causal loops linking agents, societies, and environments.

1 Introduction

One of the deepest assumptions underlying contemporary artificial intelligence is that cognition occurs primarily inside agents. Even when systems interact socially or physically, the environment is usually treated as an external substrate rather than as an active computational component. Biological evolution suggests a fundamentally different picture.

Throughout the history of life, increasing cognitive complexity has been inseparable from the ability of organisms to modify their environments and subsequently exploit those modifications as part of their adaptive dynamics. Environments become memory systems, communication channels, regulatory architectures, and sources of emergent coordination. Intelligence therefore emerges not only from brains, but from closed loops coupling agents and worlds:

$$\text{Agents} \longleftrightarrow \text{Environment}.$$

This perspective has deep roots in the history of AI and robotics. Rodney Brooks famously argued that intelligence does not arise primarily from abstract internal representations, but from situated embodied interactions with dynamic environments. LiquidAI extends this idea toward collective ecological systems, where complexity emerges from populations of agents continuously reshaping the environments that in turn shape them.

Biological evolution provides striking examples of this principle.

One evolutionary pathway is represented by ants and termites. These systems achieve planetary-scale ecological engineering despite relying on very small and relatively homogeneous brains. Their extraordinary collective intelligence emerges through self-organization, stigmergic communication, and environmental modification. Colonies construct nests, regulate temperature and humidity, optimize transportation networks, cultivate fungal agriculture, and reshape ecosystems across enormous spatial scales. In these systems, complexity is not localized within individuals but distributed across interactions and environmental structures. The colony effectively becomes a liquid cognitive architecture.

A second pathway is represented by human societies. Here, increasing cognitive complexity coevolved with larger brains, symbolic communication, and social learning. Human intelligence is fundamentally cumulative: societies store and transmit vast quantities of non-genetic information through language, institutions, technologies, and cultural systems. Human civilization can therefore be understood as a distributed cognitive ecosystem where individuals interact through layers of socially inherited external memory.

These two evolutionary trajectories suggest alternative routes toward large-scale intelligence. One route relies on massive distributed coordination among relatively simple agents coupled through environmental computation. The other relies on increasing individual cognitive complexity embedded within systems of cultural transmission and social learning. LiquidAI seeks to explore the broader space connecting these possibilities.

Our central hypothesis is that many unexplored forms of intelligence become accessible only when environments are integrated into the computational architecture itself. Once agents are allowed to modify, engineer, and computationally exploit their environments, entirely new pathways toward collective cognition and adaptive complexity may emerge.

2 Morphospaces of Possible Cognitions

A major conceptual objective of LiquidAI is the construction of *morphospaces of cognition*. In evolutionary biology, morphospaces describe the landscape of possible organismal forms, allowing researchers to distinguish realized biological structures from theoretically possible but absent ones.

We propose extending this framework to intelligence itself.

Current artificial intelligence systems occupy only a narrow region within the broader landscape of possible cognitive organizations. Biological evolution, despite its creativity, also explored only a subset of the potentially accessible architectures of intelligence.

The goal of cognitive morphospaces is therefore twofold. First, we seek to identify the fundamental dimensions underlying cognition across biological and artificial systems. Second, we aim to identify "voids" corresponding to potentially viable but unexplored forms of intelligence.

Relevant dimensions include degrees of:

- environmental coupling,
- centralization versus distribution,
- collective coordination,
- memory externalization,
- social learning,
- embodiment,
- evolutionary plasticity,
- and multiscale organization.

These morphospaces will not simply classify systems statically, but describe evolutionary trajectories through which new forms of individuality and collective organization may emerge.

3 Ecological Computation and the Evolution of Complexity

A defining principle of LiquidAI is that environments participate directly in computation.

Biological systems routinely externalize memory and coordination into environmental structures. Pheromone fields, chemical gradients, architectural structures, symbolic infrastructures, and digital communication networks all function as externalized cognitive substrates.

The key insight is that ecosystem engineering changes the computational landscape itself. As agents modify their environments, they create new informational structures and new opportunities for coordination. This generates recursive feedback loops in which:

1. agents modify environments,
2. modified environments alter interaction dynamics,
3. new forms of coordination emerge,
4. and these new collective structures enable further environmental transformation.

Such recursive loops may represent one of the fundamental engines driving the open-ended growth of biological and social complexity.

This framework also provides a new perspective on major evolutionary transitions. Increasing collective complexity is frequently associated with a redistribution of cognitive burden across scales. Ant colonies achieve remarkable adaptive sophistication using individually simple agents, whereas humans evolved increasing individual cognitive capacities tightly coupled to systems of cultural transmission and social learning.

LiquidAI seeks to understand the general principles governing these transitions and whether artificial systems can explore entirely new regions within this space.

4 Complexity Drain and Collective Organization

One of the major theoretical challenges addressed in the project concerns what we term the *complexity drain*: the tendency for complexity to migrate from individuals toward collective structures during major evolutionary transitions.

In eusocial insects, highly sophisticated colony-level organization emerges despite strong constraints on individual cognition. In human societies, by contrast, increasing collective complexity coevolved with larger brains and increasingly sophisticated social learning mechanisms.

These contrasting strategies raise fundamental questions:

- Are there universal trade-offs between individual and collective complexity?
- Under what conditions does environmental computation compensate for limited individual intelligence?
- Can hybrid AI-human systems overcome some of these limitations through multiscale cognitive architectures?

Understanding these principles is essential for building scalable collective artificial intelligence systems capable of long-term adaptive evolution.

5 Research Structure

5.1 WP1: Theory and Morphospaces of Collective Cognition

This work package develops the theoretical foundations of ecological intelligence. We will construct formal morphospaces describing possible cognitive architectures across biological and artificial systems using tools from statistical physics, information theory, evolutionary theory, and network science.

A major objective is to characterize transitions in individuality, distributed memory, social learning, and environmental coupling within collective systems. We will also develop quantitative frameworks describing complexity drain and the redistribution of adaptive functionality across scales.

5.2 WP2: Synthetic Biological Liquid Intelligence

This work package investigates the emergence of collective intelligence in engineered living systems.

Using synthetic biology, programmable multicellular assemblies, xenobots, and anthrobots, we will explore how populations of bioengineered agents can communicate, coordinate, and modify their environments. Particular emphasis will be placed on environmental coupling: agents will interact through chemical fields, mechanical modifications, and dynamically evolving local structures capable of storing information and constraining future interactions.

The objective is not merely to engineer useful biological machines, but to study whether ecological interactions among synthetic living agents can generate increasing cognitive complexity over evolutionary and developmental timescales.

These systems provide an ideal experimental substrate for investigating the emergence of collective behavior, distributed memory, adaptive morphogenesis, and transitions toward novel forms of individuality.

5.3 WP3: Digital Ecosystems and Alternative Paths to Ultrasociety

This work package explores collective intelligence within populations of interacting AI agents inhabiting spatially extended virtual ecologies.

Large language model agents will operate within environments characterized by resource constraints, communication costs, environmental persistence, and opportunities for ecosystem engineering. Agents will be able to modify their worlds, construct externalized memory structures, and evolve cooperative institutions.

A central objective is to explore alternative pathways toward ultrasociality in both purely artificial and hybrid AI–human systems. We will investigate the conditions under which large-scale cooperation emerges, stabilizes, or collapses, including scenarios involving coordination failure, information fragmentation, competition, and institutional breakdown.

Beyond these risks, the project will investigate positive tipping points in hybrid cognitive ecosystems. In economic terms, we seek to understand how AI-driven coordination mechanisms may generate increasing returns through enhanced collective problem-solving, accelerated innovation dynamics, reduced transaction costs, and improved allocation of informational and cognitive resources.

Particular attention will be devoted to identifying mechanisms through which hybrid AI–human ecologies can generate socially beneficial equilibria characterized by higher collective welfare, increased resilience, and enhanced adaptive capacity at societal scales.

5.4 WP4: Hybrid Human–AI Ecologies

The final work package investigates the emergence of mixed cognitive ecosystems involving humans, institutions, and collective AI systems.

Human societies already function as distributed cognitive structures integrating biological cognition with symbolic communication, technological infrastructures, and cultural memory. The integration of increasingly autonomous collective AI systems may fundamentally transform these dynamics.

We will study collaborative scientific discovery, distributed creativity, adaptive governance, and multiscale decision-making in hybrid ecologies where agency becomes distributed across heterogeneous populations of biological and artificial actors.

A major challenge concerns the design of resilient hybrid systems capable of maintaining trust, coordination, alignment, and long-term adaptive stability while preserving human agency and social welfare.

6 Conclusion

LiquidAI proposes a new scientific framework for understanding intelligence as an ecological and evolutionary phenomenon.

By integrating synthetic biology, artificial life, distributed AI, ecosystem engineering, and theories of collective cognition, the project seeks to move beyond the paradigm of isolated intelligent agents toward evolving ecologies of interacting minds and environments.

The ultimate ambition of LiquidAI is to explore the vast uncharted landscape of possible cognitions and to uncover forms of intelligence that emerge only when agents, societies, and environments coevolve as parts of a single adaptive system.